

```
1  import threading
2  import time
3  mutex = threading.Semaphore(1)
4  writeblock = threading.Semaphore(1)
5  read_count = 0
6  shared_data = 100
7  def reader(reader_no): 2 usages
8      global read_count
9
10     mutex.acquire()
11     read_count += 1
12
13     if read_count == 1:
14         writeblock.acquire()
15
16     mutex.release()
17
18     print(f"Reader {reader_no} is reading data = {shared_data}")
19     time.sleep(1)
20
21     mutex.acquire()
22     read_count -= 1
23
24     if read_count == 0:
25         writeblock.release()
26
27     mutex.release()
```

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27     mutex.release()
28 def writer(writer_no): 1 usage
29     global shared_data
30
31     writeblock.acquire()
32
33     shared_data += 10
34     print(f"Writer {writer_no} updated data to {shared_data}")
35     time.sleep(1)
36
37     writeblock.release()
38 r1 = threading.Thread(target=reader, args=(1,))
39 r2 = threading.Thread(target=reader, args=(2,))
40 w1 = threading.Thread(target=writer, args=(1,))
41 r1.start()
42 r2.start()
43 w1.start()
44 r1.join()
45 r2.join()
46 w1.join()
47
48 print("Execution Completed")

```

```

Reader 1 is reading data = 100
Reader 2 is reading data = 100
Writer 1 updated data to 110
Execution Completed

Process finished with exit code 0

```